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A Graph-Theoretic Analysis of the YouTube Social Network: Comparison with ER, BA, and WS Models

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[Github: https://github.com/Yumin-Hwang046/youtube-network-topology-analysis](https://github.com/Yumin-Hwang046/youtube-network-topology-analysis)

Introduction

Understanding the underlying topology of large-scale digital platforms is critical for modeling information diffusion and community formation. As one of the largest content-driven ecosystems, YouTube provides a unique environment where algorithmic content distribution intersects with complex, user-generated social structures. In this report, I analyze the structural properties of the YouTube friendship network to explore both its macro-level architectural characteristics and micro-level community dynamics.

The primary objective of this study is to map, quantify, and interpret the structural properties of this network using the analytical framework of modern network science. By abstracting the platform's user connections into mathematical graphs, I investigate whether the network topology aligns more closely with a purely random configuration, a highly centralized scale-free network, or a localized small-world structure.

Initially, it is reasonable to assume that YouTube—a platform heavily dominated by highly visible content creators acting as network hubs—might primarily follow a preferential attachment model of network growth, leading to a scale-free topology. However, modern algorithmic platforms are subject to a multitude of competing forces, meaning they rarely adhere strictly to a single theoretical generative model. While preferential attachment may explain the emergence of massive influencers, it often fails to account for the dense communities formed by everyday users interacting within niche interests.

To evaluate these dynamics, I empirically measure the network's fundamental properties: its degree distribution, clustering behavior, and path properties. I then benchmark the empirical data against three well-established theoretical null models—the Erdős–Rényi (ER) random graph, the Barabási–Albert (BA) scale-free model, and the Watts–Strogatz (WS) small-world model. Through this comparative approach, I aim to isolate the structural signatures unique to YouTube and explore whether classical generative models are sufficient to explain its structure, or if additional mechanisms—such as algorithmic recommendation systems and sociological homophily—are required to fully understand its complex hybrid topology.

Methodology

Dataset Context and Preprocessing Procedures

This study utilizes the com-youtube dataset, a large-scale network graph provided by the Stanford Network Analysis Project (SNAP). The dataset contains user-to-user friendship links, capturing the social structure of the platform during its early, community-driven operational phase. During this period, YouTube maintained a distinct reciprocal "friendship" feature, which required mutual consent to establish a connection, making it an excellent proxy for genuine bidirectional social interaction rather than merely passive content consumption (such as unidirectional subscriptions). The dataset was sourced directly from the Stanford SNAP official repository via a fixed HTTP endpoint, allowing deterministic and fully reproducible retrieval of the raw network data for this analysis.

The empirical network was mathematically constructed as an unweighted, undirected graph.

In this graph representation, $G = (V, E)$, the set of vertices V represents unique YouTube users, and the set of edges E represents a mutual friendship or connection between two individual users. The graph is unweighted, meaning that all edges are treated identically, lacking information regarding the intensity, frequency, or duration of the interaction between the connected users. Given the massive scale of the empirical graph, which encompasses over one million nodes ($N > 10^6$) and millions of edges, parsing the raw data required the implementation of highly memory-efficient data structures. Standard adjacency matrices are computationally prohibitive for graphs of this size due to their $O(N^2)$ space complexity. Consequently, sparse adjacency lists and optimized graph processing libraries were utilized to represent the network without exceeding system memory constraints.

Furthermore, exact path-based metrics—such as the exact global diameter and the average shortest path length across the entire network—require All-Pairs Shortest Path (APSP) computations. APSP algorithms typically operate with a time complexity of $O(V^3)$ or $O(VE)$, rendering them computationally intractable on a network of this immense magnitude. Thus, we adopted random sampling approaches for calculating path-dependent properties in subsequent stages. Due to the large scale of the network, certain global metrics were estimated using sampling-based approximations, ensuring computational feasibility while preserving statistical validity.

Network Metrics Evaluated

To systematically quantify the topological characteristics of the YouTube network, we calculated three fundamental network science metrics, each offering insights into different scales of the network's architecture:

1. **Degree Distribution ($P(k)$):** The degree k of a node refers to the number of edges connected to it. The degree distribution measures the probability $P(k)$ that a randomly selected node in the network has exactly k connections. In network science, the shape of this distribution is the primary diagnostic tool for assessing scale-free properties (Newman, 2010). A Poisson distribution suggests a random network, while a heavily right-skewed, heavy-tailed power-law distribution suggests a scale-free network governed by preferential attachment.
2. **Average Clustering Coefficient (C):** This local metric captures the degree of localized cliquishness or cohesiveness within the network. For a single node, its clustering coefficient is the fraction of its neighbors who are also neighbors with each other. Averaged across the entire network, it measures the systemic probability of triadic closure—the likelihood that two friends of a given user are also friends with each other. High clustering is a hallmark of strong community structure and social homophily.
3. **Average Path Length (L):** This global metric represents the average number of steps (edges) required to travel between any two randomly selected nodes along the shortest possible path. This metric helps evaluate the potential efficiency of information transfer and global navigability across the entire network space.

Null Models for Comparative Analysis

To help interpret the empirical findings and isolate unique platform-specific dynamics, the YouTube network was systematically benchmarked against three classical theoretical generative models (null models). Due to the aforementioned computational constraints associated with generating and analyzing massive theoretical graphs, these null models were scaled to a representative parameter of $N = 10,000$ nodes, carefully maintaining an average degree consistent with the empirical graph to ensure fair topological comparisons.

1. **Erdős–Rényi (ER) Model** (Erdős and Rényi, 1959): Serves as the purely random structural baseline. In the ER model, edges are formed randomly between any pair of nodes with a uniform probability p , independent of all other edges. This model generates a network with a Poisson degree distribution and negligible clustering. It is used as a baseline to assess whether the YouTube network exhibits statistically significant non-random configuration.
2. **Barabási–Albert (BA) Model** (Barabási and Albert, 1999): Represents a scale-free network generated through two simultaneous mechanisms: continuous growth (nodes are added one at a time) and preferential attachment (new nodes preferentially connect to existing nodes with high degrees). The BA model analytically yields a power-law degree distribution $P(k) \sim k^{-3}$ and is heavily utilized to model systems dominated by prominent hubs.
3. **Watts–Strogatz (WS) Model** (Watts and Strogatz, 1998): Represents a small-world network designed to bridge the gap between regular lattices and random graphs. It begins with a regular ring lattice characterized by high local clustering, and then randomly rewires edges with a tunable probability p . This rewiring process introduces long-range topological "shortcuts," drastically reducing the average path length while successfully maintaining the high local clustering inherent to the initial lattice structure.

Results: Empirical vs Theoretical

Empirical Findings

The initial exploratory analysis of the raw YouTube graph revealed a remarkably cohesive, yet highly skewed and asymmetrical topological structure. The network features an exceptionally large Largest Connected Component (LCC), encompassing the vast majority of all nodes in the dataset. The presence of such a dominant LCC indicates that the network is not highly fragmented into isolated sub-components. This suggests that despite its immense size and the diverse, decentralized nature of its user base, the platform may function as a largely unified structural entity where almost any user could theoretically reach another through a relatively short chain of mutual friendships.

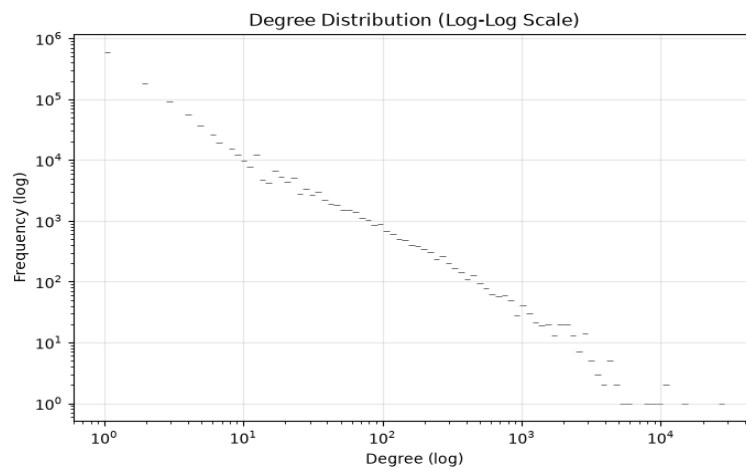


Figure1. Log-log degree distribution showing a heavy-tailed pattern.

Crucially, the empirical degree distribution is heavily right-skewed, demonstrating an extreme level of structural inequality. The vast majority of YouTube users possess only a handful of connections—often hovering near the absolute minimum required to be considered part of the active graph. In stark contrast, a minuscule fraction of outlier users possess tens of thousands of reciprocal connections. This extreme variance in node degree creates a landscape dominated by a few colossal structural anchors amidst a sea of minimally connected nodes. Such an extreme degree variance appears entirely inconsistent with a purely random formation process, heavily implying the presence of systematic bias in how connections are formed.

Comparative Analysis

When the empirical data is rigorously benchmarked against the theoretical null models, distinct and illuminating topological signatures emerge. The Erdős-Rényi (ER) model serves as our purely random structural baseline, predicting that most nodes should possess a degree very close to the network's mathematical average, producing a distinct bell-shaped probability curve. As expected from modern network theory, the empirical YouTube network exhibits substantial deviation from the ER model. It shows a pronounced heavy-tailed degree distribution rather than a Poisson-like curve, which strongly suggests that network formation on the platform is unlikely to be a purely random process driven by chance encounters.

In contrast to the ER model, the degree distribution of the empirical network shows striking structural similarity to BA-like behavior, exhibiting characteristics highly consistent with a power-law distribution. The visual alignment on a log-log plot suggests that the probability of a node having degree k roughly decays as a function of $k^{-\gamma}$, where γ is the scaling exponent. This structural alignment indicates that 'preferential attachment'—where highly connected, highly visible users (such as popular creators or established hubs) may naturally attract disproportionately more connections from new entrants—could be a highly plausible mechanism contributing to the network's macroscopic growth and hub formation.

However, an interesting and theoretically significant observation arises when analyzing the network's local cohesiveness. The pure BA model, while excellent at explaining the presence of hubs and the power-law tail, does not fully capture the relatively high clustering coefficient observed in the empirical data. In a pure BA generative process, clustering typically drops off rapidly as the network size N grows, because connections are driven solely by node degree rather than local geographic or social proximity. In the YouTube network, the empirical clustering coefficient is notably orders of magnitude higher than the BA baseline prediction.

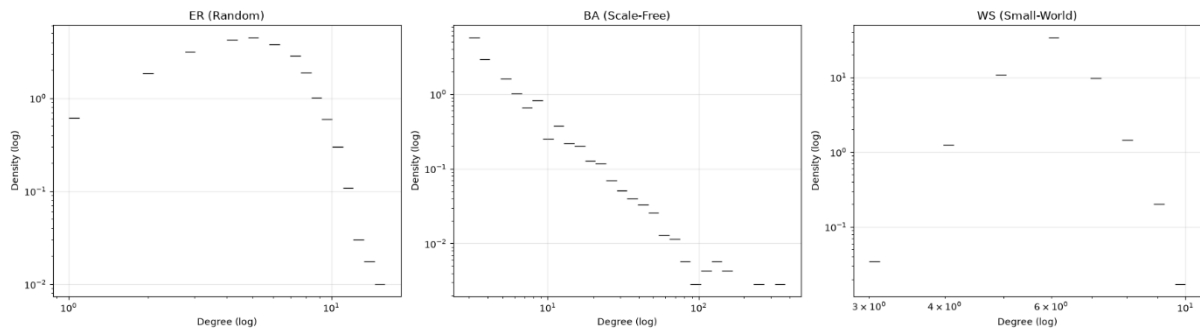


Figure2. Empirical vs ER, BA, WS degree distribution comparison.

Rather than matching the BA model's lack of local structure, the local cohesiveness of the empirical graph closely resembles that of the Watts-Strogatz (WS) small-world model, which is specifically designed to exhibit high triadic closure. This significant structural deviation may imply that while the network exhibits globally scale-free characteristics driven by popularity and visibility, it may also be microscopically composed of tightly-knit, highly redundant communities that are characteristic of organic human social circles.

Discussion: Interpreting the Deviation

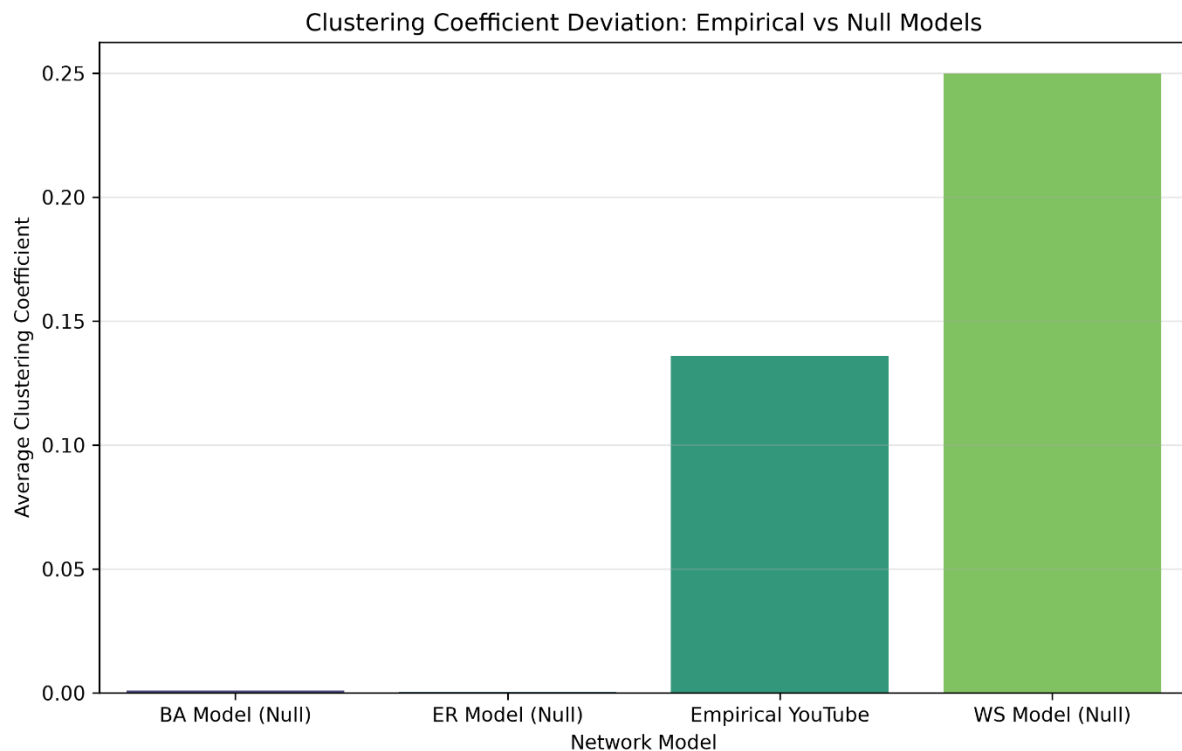


Figure3. Clustering coefficient comparison with ER, BA, and WS models.

The analytical focus can therefore shift from simply asking "Is YouTube a BA network?"—to which the answer is a qualified no—to exploring a much richer network science inquiry: "What possible mechanisms might contribute to the observed structural deviation from pure scale-free expectations?" The substantial deviation from the pure BA model regarding the clustering coefficient suggests that preferential attachment alone may not fully account for the dense, localized communities observed in the empirical data. To achieve such a hybrid topology, additional generative forces must act simultaneously on the network's evolution.

Mechanism 1: Algorithmic Influence and Triadic Closure

A commonly suggested factor in the academic literature for explaining such unnaturally high clustering in digital environments is the platform's potential algorithmic influence. Unlike biological or purely offline social networks where connections rely on physical proximity or direct chance encounters, digital social networks often actively employ sophisticated recommendation engines designed explicitly to promote user engagement and retention.

A frequent and highly effective feature of many such algorithmic engines is suggesting "friends of friends" to users. In network science terms, this specific algorithmic recommendation actively promotes and accelerates **triadic closure**. If Node A is connected to Node B, and Node B is connected to Node C, the underlying recommendation algorithm is highly likely to identify this open triad and suggest Node A to Node C, explicitly encouraging the formation of a closed structural triangle. Over millions of iterations, this type of continuous, platform-driven algorithmic reinforcement could potentially boost global local

clustering far beyond what purely random interactions or even pure preferential attachment models would typically generate on their own.

Mechanism 2: User Homophily

Another plausible explanation, seamlessly complementing potential algorithmic influence, is the well-documented sociological phenomenon of homophily (McPherson, Smith-Lovin and Cook, 2001): the fundamental human tendency of individuals to associate, bond, and form ties with similar others. In the context of a content-rich platform like YouTube, homophily is likely driven by shared semantic interests rather than mere geographic proximity.

Users may organically cluster around shared content categories, distinct niche interests, specific languages, or cultural similarities. For instance, gaming enthusiasts, beauty vloggers, independent musicians, and educational channels might organically form distinct, densely connected sub-graphs as users within these niches continually interact with, comment on, and friend one another. Such communities naturally generate a high volume of closed triads, which could significantly contribute to the elevated high clustering observed empirically. By fostering these deeply interconnected pockets of users, homophily potentially embeds local WS-like small-world structures directly within the broader, macroscopic BA-like scale-free topology. Importantly, these explanations are not mutually exclusive and may jointly contribute to the observed structure, as recommendation algorithms are often explicitly designed to exploit homophilic tendencies by grouping similar users.

Structural Vulnerability and Network Robustness

This observed hybrid topology may have profound theoretical implications for network robustness and systemic resilience, aligning closely with established theories on the error tolerance of complex networks. If the network exhibits heavy-tailed, scale-free properties at the macro level, it might be reasonably robust to random node failures. If a significantly large number of average users were to suddenly leave the platform (simulating random node removal), the overall structural integrity, the existence of the Largest Connected Component, and the global connectivity would likely remain largely unaffected. This resilience occurs because the vast majority of users possess very few connections and therefore contribute minimally to the global traversability of the network space.

Conversely, this extreme hub concentration suggests the network could be structurally fragile and highly sensitive to targeted attacks. A vanishingly small fraction of mega-hubs may control a disproportionately massive share of the network's bridging edges. If these highly connected hubs were systematically removed—either through centralized platform bans, widespread algorithmic demonetization forcing creator exodus, or migration to a competitor platform—the network might rapidly fragment into much more isolated clusters, potentially reducing the global connectivity that bridges disparate communities and severely degrading the platform's capacity for rapid information dissemination.

Analytical Limitations and Empirical Assumptions

While these topological findings provide critical insights into the platform's architecture, several methodological limitations and empirical assumptions must be explicitly acknowledged to contextualize these interpretations. First, the empirical dataset represents

a static snapshot of the network at a single, fixed point in time. Because the analysis relies on cross-sectional data, the underlying growth mechanisms—such as preferential attachment and sequential triadic closure—can only be inferred structurally rather than observed directly through longitudinal time-series tracking.

Second, the analysis inherently assumes an unweighted and undirected graph structure. Treating all edges identically masks the true heterogeneity of human relationships; it fails to differentiate between a dormant, one-time friendship link and a highly active connection characterized by frequent interaction. Similarly, while the 'friendship' feature analyzed here is reciprocal (undirected), the broader YouTube ecosystem is largely defined by asymmetric subscriber-broadcaster relationships (directed). Consequently, this analysis captures a specific subset of reciprocal social dynamics rather than the entirety of the platform's modern parasocial consumption structure. Finally, as previously noted, reliance on sampling approximations for global path metrics introduces a degree of estimation variance, necessitating appropriate statistical caution when interpreting exact global diameter values.

Conclusion

In conclusion, the empirically observed properties of the YouTube social network suggest it may not be neatly or sufficiently categorized by any single classical theoretical model. Instead, the network appears to exhibit a highly complex hybrid topology. At the macroscopic level, it reveals a structure showing strong similarities to BA-like behavior, characterized by an extreme, power-law degree distribution that is potentially influenced by visibility and preferential attachment dynamics. Simultaneously, alongside this scale-free global architecture, the network demonstrates dense local clustering akin to WS-like small-world models. This dense local structure could be plausibly associated with sociological user homophily and the systemic acceleration of algorithmic triadic closure (e.g., automated 'friends of friends' recommendations).

The empirical alignment with the power-law degree distribution is highly consistent with the theoretical idea that popularity, fame, and visibility may play a disproportionate role in macroscopic network growth. Conversely, the structural deviation in the clustering coefficient compared to purely scale-free models suggests the possible, robust presence of denser, more insular communities operating beneath the overarching hub structure. This potential hybrid nature implies an asymmetric resilience profile: the network could be reasonably robust to random user drop-offs and everyday churn, yet might remain structurally brittle and highly sensitive to targeted attacks on its central, highly connected hubs.

From a broader academic perspective, these empirical observations suggest that classical preferential attachment alone may be insufficient to fully model the architectural complexities of modern, algorithmically-mediated social networks. Since the results presented herein are primarily observational and derived from a static abstraction, future research should proactively look toward expanding this static, undirected analytical framework. Incorporating edge weights to measure interaction intensity and frequency, mapping directional links to accurately capture asymmetric subscriber versus publisher dynamics, and leveraging longitudinal time-series data to directly observe temporal network evolution would be highly valuable, necessary steps toward empirically testing and validating the mechanistic theories of algorithmic influence and homophily discussed in this report.

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